DATA MINING & BUSINESS INTELLIGENCE

Laboratory Experiment Record

EXPERIMENT NO. 4

To Implement Classification Algorithms (Decision Tree and Naïve Bayes) using RapidMiner — Airline Passenger Satisfaction

Student Name	_____
Roll No. / Enrollment No.	_____
Class / Semester	_____
Date of Performance	_____
Faculty Signature	_____

1. Aim

To implement the Decision Tree and Naïve Bayes classification algorithms on an airline dataset using the RapidMiner (Altair AI Studio) tool, and to compare their performance using accuracy, precision, recall and the confusion matrix.

2. Theory

2.1 Classification

Classification is a supervised data mining technique. It learns from labelled records (the training set) and then predicts the class label of new, unseen records (the test set). In this experiment the class label is 'satisfaction', and the task is to predict whether an airline passenger is 'satisfied' or 'neutral or dissatisfied'.

2.2 Decision Tree

A decision tree repeatedly splits the data on the attribute that best separates the classes. Each internal node tests an attribute, each branch is an outcome of that test, and each leaf gives a class label. The result is a set of easy-to-read if-then rules. Trees are simple to interpret but can overfit if they grow too deep.

2.3 Naïve Bayes

Naïve Bayes is a probabilistic classifier based on Bayes' theorem, $P(C|X) = P(X|C) \times P(C) / P(X)$. It assumes that all attributes are independent of each other once the class is known (the 'naïve' assumption). It is fast and works well on large datasets, but the independence assumption is rarely perfectly true.

2.4 Evaluation Measures

Measure	Meaning
Accuracy	Percentage of test records classified correctly.
Confusion matrix	Table of predicted classes against actual classes.
Precision	Of the records predicted as a class, how many really belong to it.
Recall	Of the records that really belong to a class, how many were found.
F-measure	Harmonic mean of precision and recall.

3. Tool and Dataset

Tool: RapidMiner (Altair AI Studio, Educational license).

Dataset: Airline Passenger Satisfaction (Kaggle), file train.csv, with 103,904 records. It holds passenger details, flight details and 1–5 service ratings, and 'satisfaction' is the class label.

Source: kaggle.com/datasets/teejmahal20/airline-passenger-satisfaction

Attribute Group	Attributes	Type
Passenger info	Gender, Customer Type, Age	Nominal / Numeric
Flight info	Type of Travel, Class, Flight Distance, Departure and Arrival Delay	Nominal / Numeric

Attribute Group	Attributes	Type
Service ratings (1–5)	Inflight wifi service, Online boarding, Seat comfort, Inflight entertainment, Cleanliness, Baggage handling, etc.	Numeric
Class label (target)	satisfaction: 'satisfied' or 'neutral or dissatisfied'	Nominal

The data was split 70% for training and 30% for testing. The test partition therefore holds 31,171 records, which is the total of each confusion matrix shown in Section 5.

4. Procedure

4.1 Data Preparation (common to both algorithms)

1. Open RapidMiner, create a Blank Process, and drag the imported dataset (Retrieve) onto the canvas.
2. Add Select Attributes (type = exclude attributes) to remove the id column, the unnamed index column and low-importance attributes such as Gender and Gate location.
3. Add Replace Missing Values to fill the blank Arrival Delay values.
4. Add Set Role and set the attribute 'satisfaction' as the label.
5. Add Split Data with partitions 0.7 and 0.3 and shuffled sampling.

4.2 Decision Tree

1. Add the Decision Tree operator and connect the first partition (70%) of Split Data to its training input.
2. Add Apply Model. Connect the tree's model output to its model input, and the second partition (30%) of Split Data to its unlabelled input.
3. Add Performance (Classification), tick accuracy, precision and recall, and connect Apply Model's labelled output to it.
4. Connect the performance output and the model to the results ports and click Run.

4.3 Naïve Bayes

1. Replace the Decision Tree operator with Naive Bayes, keeping every connection the same.
2. Click Run, then open the Description, Plot view and Distribution Table of the model, and the Performance Vector.

5. Implementation and Output

5.1 Decision Tree

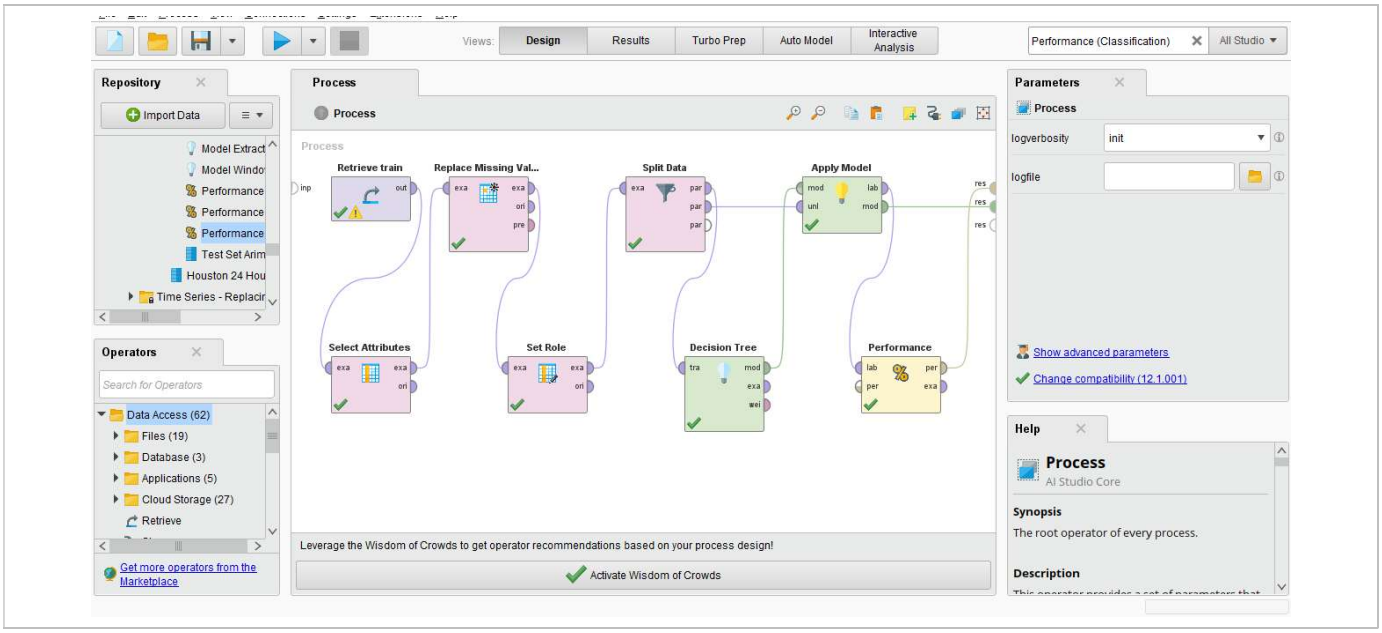
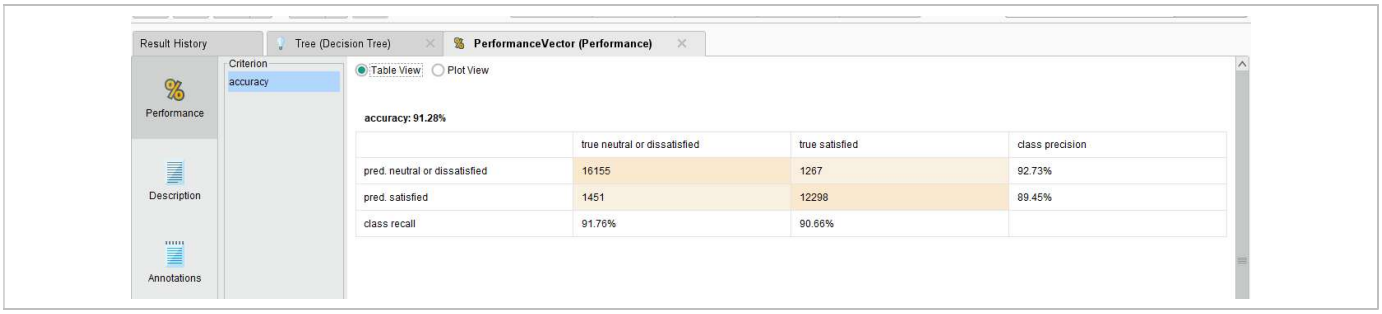


Figure 1: Decision Tree process in RapidMiner (Retrieve → Select Attributes → Replace Missing Values → Set Role → Split Data → Decision Tree → Apply Model → Performance)



	true neutral or dissatisfied	true satisfied	class precision
pred. neutral or dissatisfied	16155	1267	92.73%
pred. satisfied	1451	12298	89.45%
class recall	91.76%	90.66%	

Figure 2: Decision Tree performance — accuracy 91.28% with confusion matrix

The Decision Tree classified 91.28% of the 31,171 test records correctly. It correctly predicted 16,155 'neutral or dissatisfied' and 12,298 'satisfied' passengers, and made 2,718 errors (8.72%).

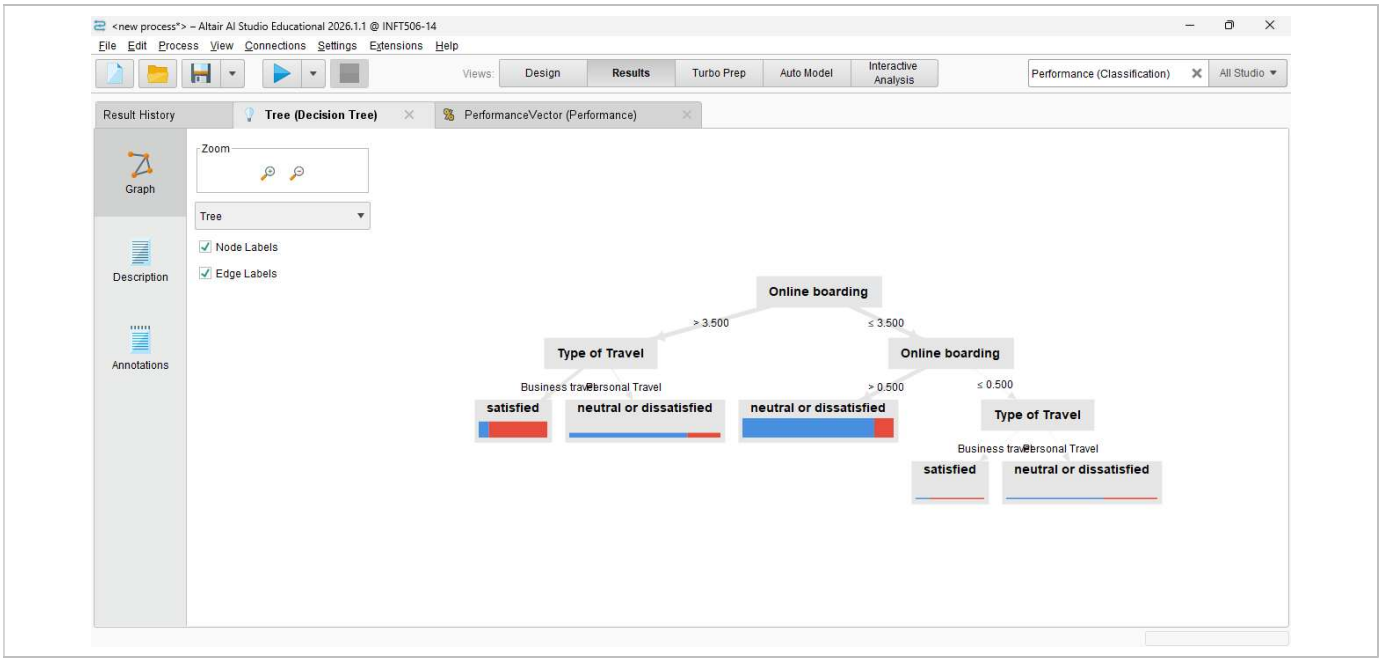


Figure 3: Decision Tree (Graph view)

The tree uses only two attributes. The root node splits on Online boarding at 3.5. On the high side (above 3.5), Type of Travel decides the outcome: business travellers are classified as 'satisfied' and personal travellers as 'neutral or dissatisfied'. On the low side, most passengers are 'neutral or dissatisfied', apart from those with a very low online boarding score (0.5 or below) who are business travellers and are classified as 'satisfied'.

5.2 Naïve Bayes

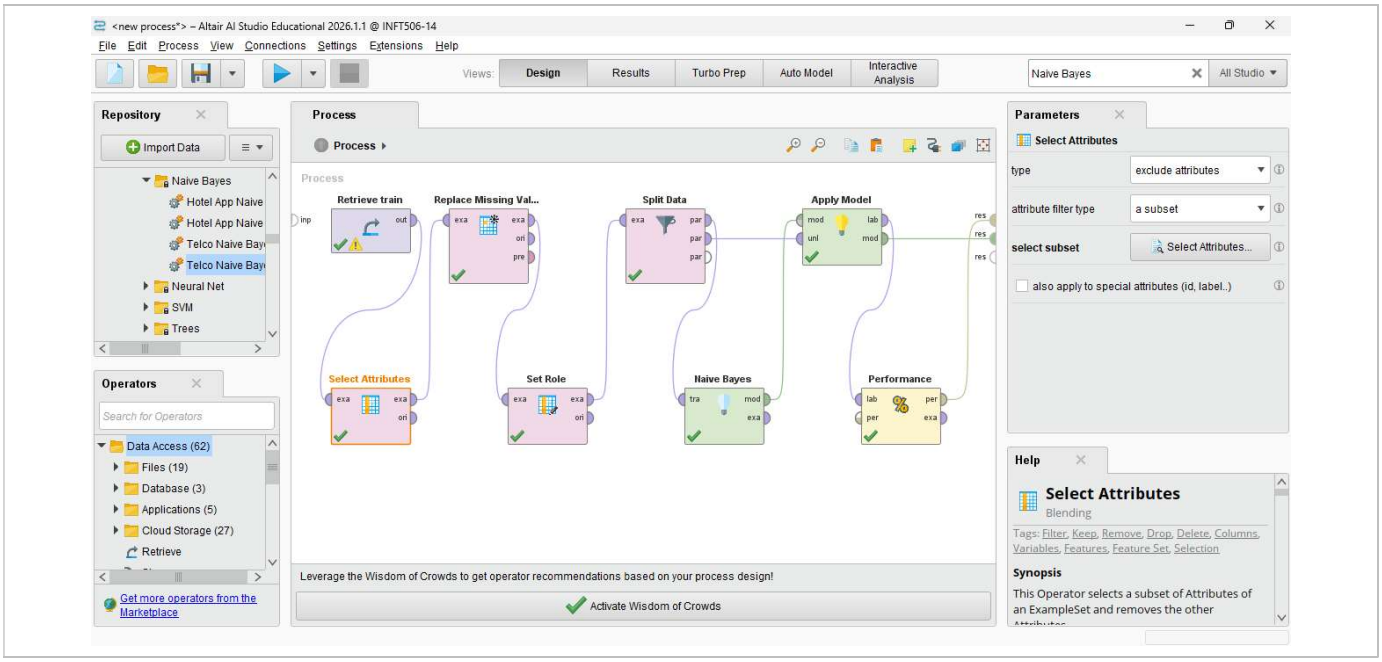


Figure 4: Naïve Bayes process in RapidMiner (the Decision Tree operator replaced by Naive Bayes)

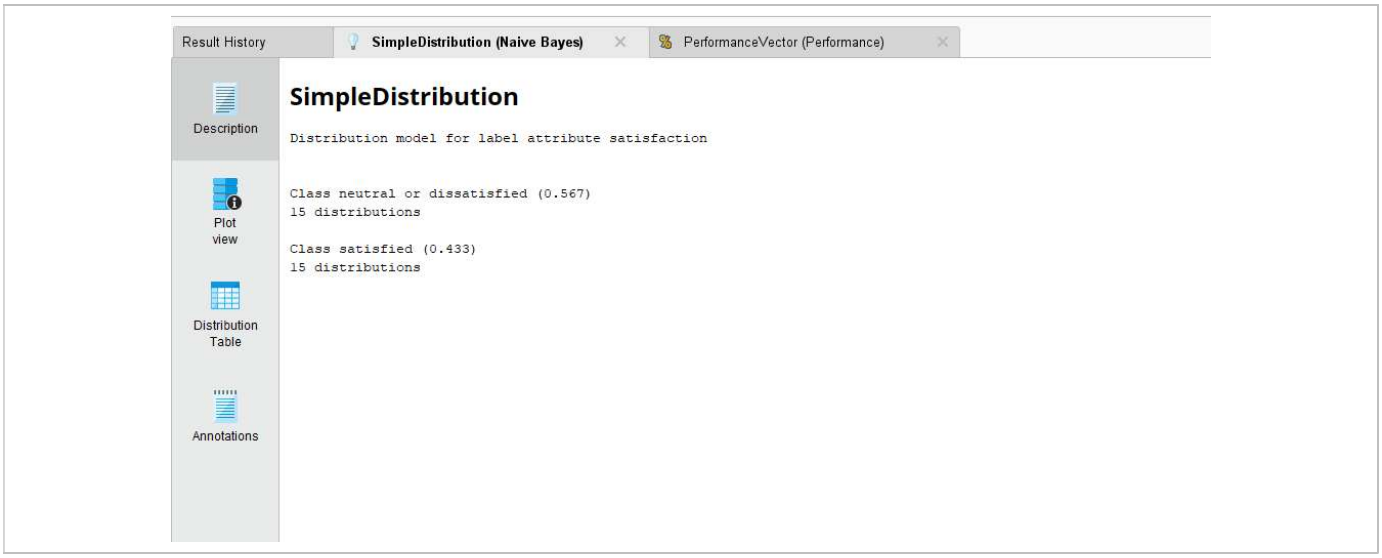


Figure 5: Naïve Bayes model description

The description shows the prior probabilities of the two classes: 56.7% 'neutral or dissatisfied' and 43.3% 'satisfied'. The model built 15 distributions per class, one for each attribute that was kept.

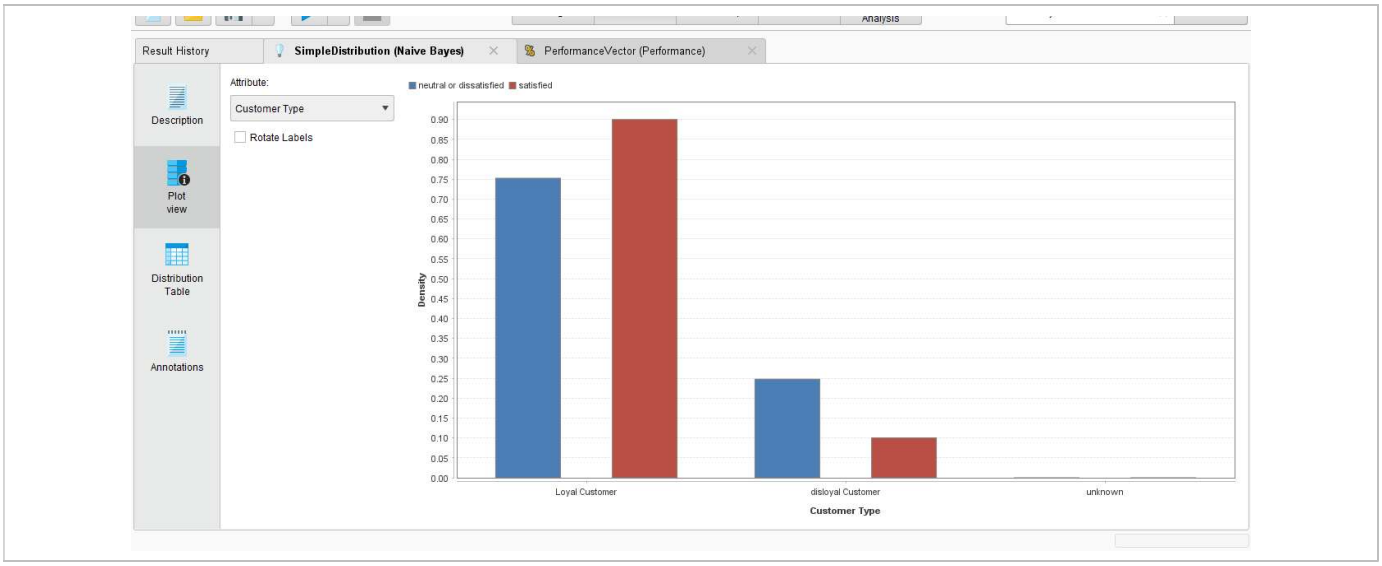
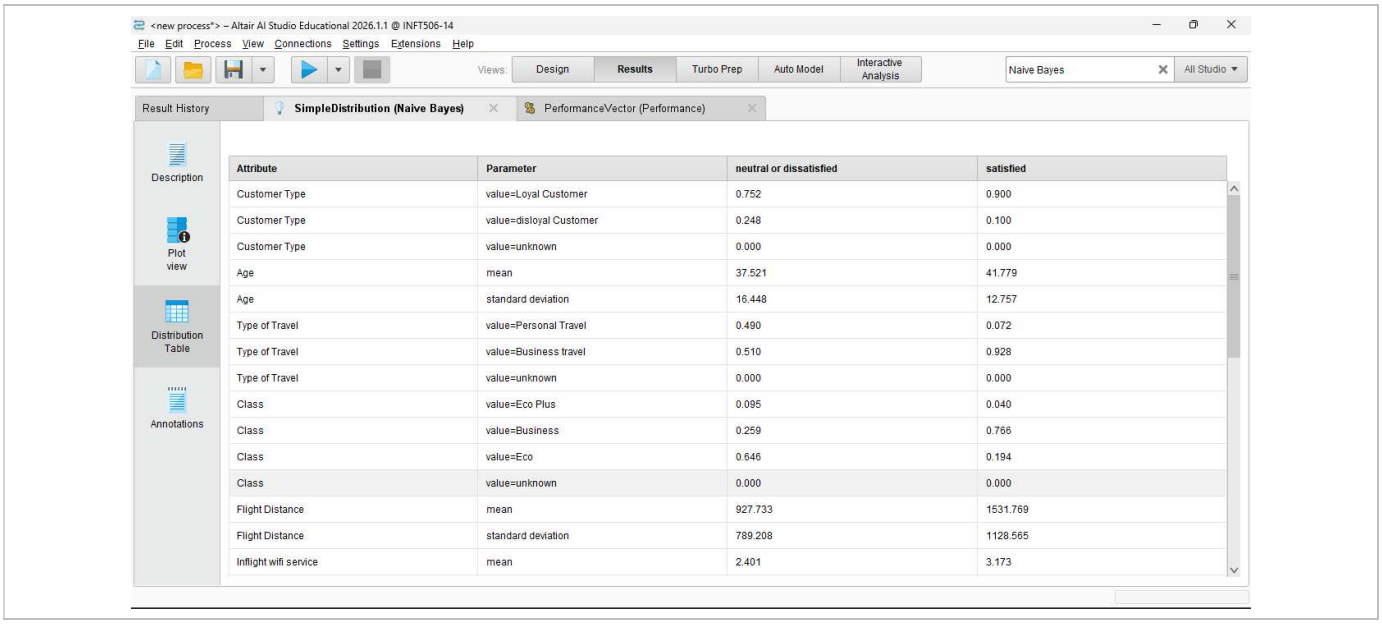


Figure 6: Naïve Bayes Plot view (attribute: Customer Type)

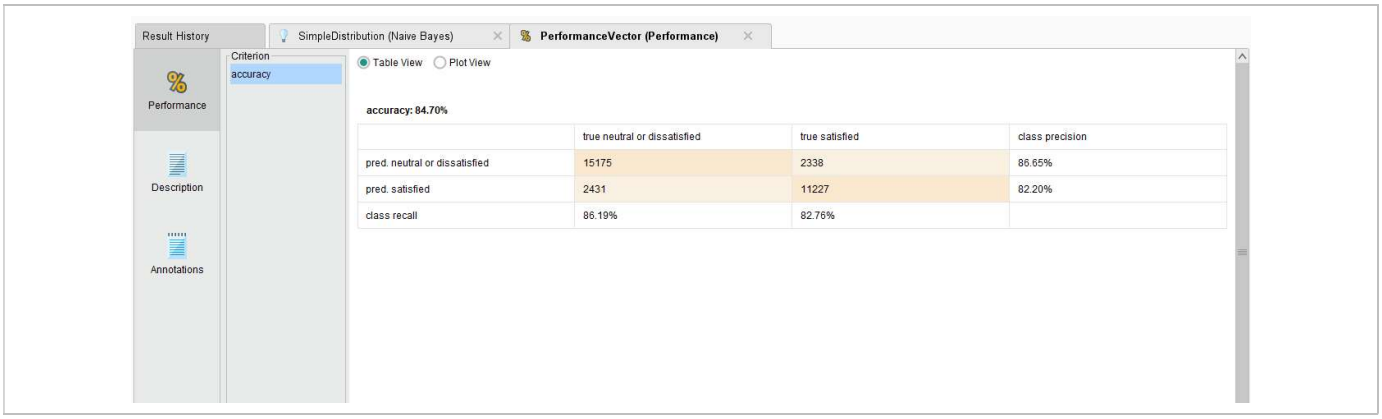
For Customer Type, loyal customers are more common among satisfied passengers (density about 0.90) than among neutral or dissatisfied passengers (about 0.75), while disloyal customers show the opposite pattern.



Attribute	Parameter	neutral or dissatisfied	satisfied
Customer Type	value=Loyal Customer	0.752	0.900
Customer Type	value=disloyal Customer	0.248	0.100
Customer Type	value=unknown	0.000	0.000
Age	mean	37.521	41.779
Age	standard deviation	16.448	12.757
Type of Travel	value=Personal Travel	0.490	0.072
Type of Travel	value=Business travel	0.510	0.928
Type of Travel	value=unknown	0.000	0.000
Class	value=Eco Plus	0.095	0.040
Class	value=Business	0.259	0.766
Class	value=Eco	0.646	0.194
Class	value=unknown	0.000	0.000
Flight Distance	mean	927.733	1531.769
Flight Distance	standard deviation	789.208	1128.565
Inflight wifi service	mean	2.401	3.173

Figure 7: Naïve Bayes Distribution Table

The distribution table gives the probabilities and means that Naïve Bayes uses to classify each record. For example, 92.8% of satisfied passengers travelled on business, against 51.0% of neutral or dissatisfied passengers. Satisfied passengers also had a higher mean Flight Distance (about 1,532 against 928) and a higher mean Inflight wifi rating (3.17 against 2.40).



	true neutral or dissatisfied	true satisfied	class precision
pred. neutral or dissatisfied	15175	2338	86.65%
pred. satisfied	2431	11227	82.20%
class recall	86.19%	82.76%	

Figure 8: Naïve Bayes performance — accuracy 84.70% with confusion matrix

Naïve Bayes classified 84.70% of the test records correctly, with 4,769 errors (15.30%).

6. Result Comparison

Measure	Decision Tree	Naïve Bayes
Accuracy	91.28%	84.70%
Misclassified records	2,718 (8.72%)	4,769 (15.30%)
Precision — neutral or dissatisfied	92.73%	86.65%
Precision — satisfied	89.45%	82.20%
Recall — neutral or dissatisfied	91.76%	86.19%
Recall — satisfied	90.66%	82.76%
F-measure — neutral or dissatisfied *	92.24%	86.42%
F-measure — satisfied *	90.05%	82.48%
Test records	31,171	31,171

* F-measure calculated from the precision and recall values in the performance tables.

Confusion matrices

Decision Tree	True neutral or dissatisfied	True satisfied
Predicted neutral or dissatisfied	16,155	1,267
Predicted satisfied	1,451	12,298

Naïve Bayes	True neutral or dissatisfied	True satisfied
Predicted neutral or dissatisfied	15,175	2,338
Predicted satisfied	2,431	11,227

7. Conclusion

The Decision Tree and Naïve Bayes classifiers were implemented in RapidMiner on the Airline Passenger Satisfaction dataset to predict whether a passenger is satisfied. The Decision Tree achieved an accuracy of 91.28% and Naïve Bayes achieved 84.70% on the same 30% test partition, and the Decision Tree was also better on precision, recall and F-measure for both classes.

The Decision Tree also gave clear, readable rules, showing that Online boarding and Type of Travel are the most important factors for passenger satisfaction. Naïve Bayes was simpler and gave useful class probabilities, but its accuracy is lower because the attributes (for example, the different service ratings) are not truly independent of each other. For this dataset, the Decision Tree is therefore the better classifier in both accuracy and interpretability.